Layer 0: Wound Biology

Protease Cascade ODE: MMP-9 and HNE in Chronic Wound Fluid

Eton iGEM 2026, Dry Lab | Layer 0 of 17-Layer Biosensor Modelling Pipeline | `wound_biology.py`

Purpose

Layer 0 predicts the **free protease concentration in wound fluid at the time of swab sampling** and how it evolves over the 2-hour post-sampling window. Its outputs, $[\text{MMP-9}]_{\text{free}}(t)$ and $[\text{HNE}]_{\text{free}}(t)$, are in-situ wound fluid concentrations. Layer 1 (sample collection) applies the swab dilution factor D_f to convert from in-situ concentrations to what enters the device.

The distinction between healing and chronic wound states is encoded through two distinct initial-condition parameter sets, both grounded in published ELISA data. The model is a system of ODEs for nine coupled species: pro-MMP-9, free MMP-9, free TIMP-1, MMP-9:TIMP-1 complex, free HNE, free A1AT, HNE:A1AT complex, free SLPI, and HNE:SLPI complex.

Parameter citation policy. Every parameter used by this model must be grounded in a paper in `electrochemical_device/papers/layer_00/` (status: ✓), approximate from a figure in such a paper (status: ∼), or explicitly flagged as awaiting measurement (status: ○). No values from secondary literature alone.

I. Wound Fluid Composition

Biological rationale

Chronic wounds are characterised by a prolonged inflammatory phase driven by persistent neutrophil and macrophage infiltration[1]. This sustains elevated protease secretion long after acute wounds have resolved. The two proteases targeted by this device are:

- **MMP-9 (gelatinase B)**: secreted by neutrophils and macrophages as an inactive zymogen (pro-MMP-9). Activation requires proteolytic removal of the prodomain. In chronic wounds, both forms are elevated; the ratio of active to pro form predicts healing outcome.
- **HNE (human neutrophil elastase)**: a serine protease released already active from azurophil granules. Regulated primarily by α_1-antitrypsin (A1AT) and secondarily by SLPI.

Literature evidence

Trengove et al. (1999)[1] measured protease activity in wound fluid from chronic venous leg ulcers (VLU) versus acute wounds (all values from paper text, p.446–448):

- Chronic VLU MMP activity: median 22.8 μg/mL, versus 0.76 μg/mL in acute wounds ($\approx 30\times$ elevation, $p < 0.001$; 92-kDa MMP-9 predominant on zymography).
- TIMP-1 (ELISA): chronic VLU median 0.8 μg/mL = 800 ng/mL ($n = 5$; text p.448).
- HNE (activity assay): non-healing VLU median 1.4 μg/mL $\approx$ 47 nM (range < 1–11.9 μg/mL; elevated in only 6 of 13 chronic samples; text p.448).
- MMP activity decreased significantly as wounds healed ($p < 0.01$; text p.449).

Ladwig et al. (2002)[2] used quantitative gelatin zymography and ELISA in wound fluid from 56 patients with chronic pressure ulcers. Key values (all from paper text p.31):

- Pro-MMP-9: mean 8,100 ± 3,000 ng/mL. Active MMP-9: mean 4,800 ± 1,300 ng/mL.
- Total MMP-9 = 8,100 + 4,800 = 12,900 ng/mL; pro-form fraction = 8,100/12,900 = 63%.
- Active MMP-9 significantly higher in poor healers at Day 0 (text p.33; approximate values from Figure 8).
- TIMP-1 significantly higher in good healers at Day 0 (text p.33; approximate values from Figure 10).
- The MMP-9:TIMP-1 ratio was a predictor of healing trajectory (text p.34, Figure 11).

Outstanding measurements:

Healing wound concentrations (not confirmed): Ladwig 2002 is a pressure ulcer study; the healing scenario MMP-9 ($\approx$ 5,400 ng/mL) is read approximately from Figure 8 (not stated in text). The healing TIMP-1 (= 1,104 ng/mL) has no paper-text source – it is a working estimate. The healing wound prediction (Section V) is therefore illustrative. It depends on the sign of the MMP-9/TIMP-1 excess, not the exact values; a confirmed joint VLU ELISA (MMP-9 + TIMP-1 in the same cohort, chronic and healing time points) is needed before the discrimination ratio is finalised.

Initial conditions

Status: ✓ confirmed from paper text, ∼ approximate from figure, ∘ needs measurement.

Parameter	Chronic wound	Healing wound	St.	Source
$[\text{MMP-9}]_{\text{total}}$	12,900 ng/mL (140 nM)	5,400 ng/mL (59 nM)	✓/∼	[2] text/Fig. 8
[pro-MMP-9] (63%)	8,100 ng/mL (88 nM)	3,400 ng/mL (37 nM)	✓/∼	[2] text
[active MMP-9] (37%)	4,800 ng/mL (52 nM)	2,000 ng/mL (22 nM)	✓/∼	[2] text/Fig. 8
[TIMP-1]	800 ng/mL (28 nM)	1,104 ng/mL (39 nM)	✓/∘	[1] text/estimate
[HNE] (both states)	47 nM (1.4 μg/mL)		✓	[1] text
[A1AT] (both states)	1.92 μM (estimate)		∘	serum ÷10

Physical basis for wound-state discrimination. In chronic wounds, active MMP-9 (52 nM) exceeds TIMP-1 (28 nM) by 24 nM. Since $K_i(\text{TIMP-1/MMP-9}) = 8.5$ nM (Olson 1997[4], confirmed), which is small compared to both concentrations, TIMP-1 is nearly fully bound and free MMP-9 persists at ≈ 32 nM (ODE result, t = 30 min). In healing wounds (estimated), TIMP-1 (39 nM) exceeds active MMP-9 (22 nM) by 17 nM, suppressing free MMP-9 to ≈ 6 nM.

II. Activation Kinetics (MMP-9 Pathway)

Pro-MMP-9 activation by MMP-3 and plasmin is modelled as pseudo-first-order in pro-MMP-9:

$$\frac{d[\text{pro-MMP-9}]}{dt} = -\left(\frac{k_{\text{cat}}}{K_m}\right)_{\text{MMP3}}[\text{MMP-3}]\,[\text{pro-MMP-9}] - \left(\frac{k_{\text{cat}}}{K_m}\right)_{\text{plasmin}}[\text{plasmin}]\,[\text{pro-MMP-9}] \tag{1}$$

Outstanding measurements:

$(k_{\text{cat}}/K_m)_{\text{MMP3}}$, $(k_{\text{cat}}/K_m)_{\text{plasmin}}$, [MMP-3], and [plasmin] in wound fluid are all unconfirmed placeholders. No paper in the papers folder provides wound-fluid MMP-3 or plasmin concentrations or the relevant kinetic constants. With current placeholder values, $< 1\%$ of pro-MMP-9 is activated in 120 min, so these values have no effect on the chronic wound free MMP-9 prediction. They would matter only if pro-MMP-9 activation within the 2-hour window were a significant fraction of the total.

III. Inhibition Kinetics

TIMP-1 / MMP-9 (reversible tight inhibition)

TIMP-1 binds to the MMP-9 active site in a non-covalent, reversible manner with biphasic kinetics (two binding sites; high-affinity site dominates at wound fluid concentrations):

$$\text{MMP-9} + \text{TIMP-1} \underset{k_{\text{off}}}{\overset{k_{\text{on}}}{\rightleftharpoons}} \text{MMP-9:TIMP-1} \qquad K_i = \frac{k_{\text{off}}}{k_{\text{on}}} \tag{2}$$

Confirmed kinetic constants (Olson et al. 1997[4], Table IV, enzyme inhibition assay, MMP-9 82 kDa active form):

$$k_{\text{on}} = 2.48 \pm 0.47 \times 10^5\ \text{M}^{-1}\text{s}^{-1}$$
$$k_{\text{off}} = 2.1 \pm 1.1 \times 10^{-3}\ \text{s}^{-1}$$
$$K_i = k_{\text{off}}/k_{\text{on}} = 8.5 \pm 2.9\ \text{nM} \quad \text{(stated directly in Table IV; check: } 2.1 \times 10^{-3}/2.48 \times 10^5 = 8.47\ \text{nM)}$$

Enzyme inhibition assay values are used in preference to SPR values (Olson 1997 Tables I and II) because they

reflect solution-phase kinetics at conditions closer to wound fluid. As noted by Olson 1997 (p.29981), enzyme inhibition k_{on} values are consistently 2.5–7× higher than SPR values, making them more appropriate for modelling diffusion-limited association in wound fluid.

Sensitivity of discrimination ratio to K_i. The chronic wound prediction (≈ 32 nM free MMP-9 at 30 min) is robust: changing K_i from 8.5 nM to 1 nM shifts the result from 32 nM to 27 nM (a 15% change). The healing prediction is more sensitive: $K_i = 8.5$ nM gives 6 nM free MMP-9, whereas $K_i = 1$ nM gives 1.3 nM. This is because the healing wound MMP-9/TIMP-1 excess is only 17 nM, comparable to K_i.

A1AT / HNE (near-irreversible inhibition)

A1AT forms a covalent acyl-enzyme intermediate with HNE that is effectively irreversible:

$$\text{HNE} + \text{A1AT} \xrightarrow{k_{on}} \text{HNE:A1AT} \quad \text{(irreversible; } k_{off} \approx 0\text{)} \tag{3}$$

Confirmed kinetic constant (Beatty, Bieth & Travis 1980[5], Table 1):

$$k_{on}(\text{HNE/A1AT}) = 6.5 \pm 4.0 \times 10^7 \text{ M}^{-1}\text{s}^{-1} \quad \text{(at 25 °C, pH 8.0)} \tag{4}$$

Half-life of free HNE (with $[\text{A1AT}]_0 = 1.92$ μM, estimated):

$$t_{1/2}^{\text{HNE}} = \frac{\ln 2}{k_{on} \cdot [\text{A1AT}]_0} = \frac{0.693}{6.5 \times 10^7 \times 1.92 \times 10^{-6}} = 0.0055 \text{ s} \quad (5.5 \text{ ms}) \tag{5}$$

Because $[\text{A1AT}]_0$ is an estimate, the conclusion is verified at 10× lower $[\text{A1AT}]_0$: $t_{1/2} = 55$ ms, still three orders of magnitude below the 5-minute measurement window.

Critical finding:

HNE channel is not viable. Free HNE has a half-life of ∼5.5 ms under estimated wound conditions (confirmed $k_{on} = 6.5 \times 10^7$ $\text{M}^{-1}\text{s}^{-1}$, Beatty 1980[5]). ODE integration confirms $[\text{HNE}]_{\text{free}} = 0.00$ pM at all $t \geq 5$ min in both wound states. The conclusion is insensitive to $[\text{A1AT}]_0$ uncertainty: even at 10× lower A1AT than assumed, $t_{1/2} = 55$ ms. The second electrode must serve as an internal reference or control, not a HNE sensing channel.

SLPI / HNE (secondary inhibitor)

SLPI is a serine protease inhibitor expressed in wound healing tissue[6]. Placeholder kinetic constants ($k_{on} = 10^5$ $\text{M}^{-1}\text{s}^{-1}$, $k_{off} = 10^{-4}$ s^{-1}) are used. Ashcroft et al. (2000)[6] confirmed the importance of SLPI to wound healing outcomes in knockout mice but did not state a quantitative SLPI concentration in human wound fluid. Because HNE is already completely neutralised by A1AT, the SLPI sub-system does not influence any device-relevant prediction.

IV. Full ODE System

Variable key (all concentrations in mol/L):

P	pro-MMP-9	C_1	MMP-9:TIMP-1 complex
M	free (active) MMP-9	H	free HNE
T	free TIMP-1	A	free A1AT
C_2	HNE:A1AT complex	S	free SLPI
C_3	HNE:SLPI complex		

$$\dot{P} = -k^*\, P \tag{6}$$

$$\dot{M} = +k^*\, P - k^T_{\text{on}} MT + k^T_{\text{off}} C_1 \tag{7}$$

$$\dot{T} = -k^T_{\text{on}} MT + k^T_{\text{off}} C_1 \tag{8}$$

$$\dot{C}_1 = +k^T_{\text{on}} MT - k^T_{\text{off}} C_1 \tag{9}$$

$$\dot{H} = -k^A_{\text{on}} HA - k^S_{\text{on}} HS + k^S_{\text{off}} C_3 \tag{10}$$

$$\dot{A} = -k^A_{\text{on}} HA \tag{11}$$

$$\dot{C}_2 = +k^A_{\text{on}} HA \tag{12}$$

$$\dot{S} = -k^S_{\text{on}} HS + k^S_{\text{off}} C_3 \tag{13}$$

$$\dot{C}_3 = +k^S_{\text{on}} HS - k^S_{\text{off}} C_3 \tag{14}$$

where $k^* = (k_{\text{cat}}/K_m)_{\text{MMP3}}[\text{MMP-3}] + (k_{\text{cat}}/K_m)_{\text{plasmin}}[\text{plasmin}]$; $k^T_{\text{on,off}}$ for TIMP-1/MMP-9 (Olson 1997[4]); k^A_{on} for A1AT/HNE (Beatty 1980[5]).

Solved with the Radau stiff solver (`scipy.integrate.solve_ivp`, `rtol=1e-10, atol=1e-15`). Stiff integration is required because the A1AT/HNE timescale (∼5.5 ms) is 10^7-fold faster than the pro-MMP-9 activation timescale (∼30–120 min). The TIMP-1/MMP-9 equilibration is intermediate ($\tau \approx 2$ min).

V. Model Output and Wound-State Discrimination

Discrimination between wound states at $t = 30$ min

Parameter	Chronic wound	Healing wound	C/H	Source
$[\text{MMP-9}]_{\text{total}}$ (in-situ)	12,900 ng/mL (140 nM)	5,400 ng/mL (59 nM)	2.4×	[2] text/Fig.
$[\text{MMP-9}]_{\text{active}}$ (in-situ)	4,800 ng/mL (52 nM)	2,000 ng/mL (22 nM)	2.4×	[2] text/Fig.
[TIMP-1] (in-situ)	800 ng/mL (28 nM)	1,104 ng/mL (39 nM)	0.72×	[1] text/est.
MMP-9 vs TIMP-1 excess	+24 nM (MMP-9)	−17 nM (TIMP-1)	–	derived
ODE output at $t = 30$ min (in-situ, before Layer 1 dilution D_f):				
$[\text{MMP-9}]_{\text{free}}$	**32** nM	**6** nM	**5**×	ODE
$[\text{TIMP-1}]_{\text{free}}$	≈6 nM	≈22 nM	–	ODE
$[\text{HNE}]_{\text{free}}$	0.00 pM	0.00 pM	n/a	ODE

Confirmed chronic wound prediction. With Ladwig 2002 text-confirmed concentrations (active MMP-9 = 52 nM, TIMP-1 = 28 nM)[2] and Olson 1997 confirmed kinetics ($K_i = 8.5$ nM)[4], the ODE gives free MMP-9 ≈ 32 nM at $t = 30$ min in chronic wounds. This is driven entirely by the 24 nM MMP-9 excess over TIMP-1; no production rate assumptions are needed.

Illustrative healing comparison. The healing result (≈ 6 nM) depends on unconfirmed estimates: healing MMP-9 from Ladwig Figure 8 (approximate) and healing TIMP-1 with no paper source. The discrimination ratio (∼5×) should be treated as a working estimate until a joint VLU ELISA is performed. The mechanism is sound: when TIMP-1 excess exceeds active MMP-9, free MMP-9 is suppressed; the exact ratio follows from the confirmed K_i.

Layer 1 dilution and sensor range

For chronic wound free MMP-9 (32 nM) to land in the sensor range (17–200 ng/mL = 0.185–2.17 nM):

$$D_f = \frac{[\text{MMP-9}]_{\text{free, in-situ}}}{[\text{MMP-9}]_{\text{target}}} = \frac{32\text{ nM}}{0.185\text{–}2.17\text{ nM}} \approx 15\text{–}173 \tag{15}$$

A dilution factor $D_f \approx 10$–100 is within the range of practical swab designs. Example with $D_f = 50$: chronic wound device entry $= 32/50 = 0.64$ nM $= 59$ ng/mL (within sensor range); healing wound $= 6/50 = 0.12$ nM $= 11$ ng/mL (near the lower end). The healing signal approaching the LOD under large D_f is a risk to address in Layer 1.

VI. Parameter Summary

Status: ✓ confirmed from paper text, ∼ from figure (not text), ∘ needs measurement/citation. **Sensitivity**: whether changing the value by 10× affects the device-relevant predictions.

Parameter	Value	Source	St.	Affects	Sensitv.
$[\text{MMP-9}]_{\text{pro}}$ (chr.)	8,100 ng/mL	Ladwig 2002[2] text p.31	✓	I.C.	Low
$[\text{MMP-9}]_{\text{act}}$ (chr.)	4,800 ng/mL	Ladwig 2002[2] text p.31	✓	I.C.	High
Pro-form fraction	63%	Ladwig 2002[2] (8100/12900)	✓	I.C.	Low
[TIMP-1] (chronic)	800 ng/mL	Trengove 1999[1] text p.448	✓	MMP-9 eq.	High
[HNE]	47 nM	Trengove 1999[1] text p.448	✓	HNE (n/v)	n/a
$[\text{MMP-9}]_{\text{act}}$ (heal.)	2,000 ng/mL	Ladwig 2002[2] Fig. 8 (approx.)	∼	I.C. (heal.)	High
[TIMP-1] (healing)	1,104 ng/mL	No paper in folder – estimate	∘	MMP-9 eq.	High
k_{on} (TIMP-1/MMP-9)	2.48×10^5 $\text{M}^{-1}\text{s}^{-1}$	Olson 1997[4] Table IV	✓	MMP-9 eq.	Low
k_{off} (TIMP-1/MMP-9)	2.1×10^{-3} s^{-1}	Olson 1997[4] Table IV	✓	K_i	Moderate
K_i (TIMP-1/MMP-9)	8.5 nM	Olson 1997[4] Table IV	✓	MMP-9 eq.	Moderate
k_{on} (A1AT/HNE)	6.5×10^7 $\text{M}^{-1}\text{s}^{-1}$	Beatty 1980[5] Table 1	✓	HNE n/v	Low
[A1AT]	1.92 μM	Serum ÷10 estimate	∘	HNE $t_{1/2}$	Low*
[MMP-3]	10 ng/mL	Placeholder	∘	Act. rate	None
[plasmin]	1 nM	Placeholder	∘	Act. rate	None
k_{cat}/K_m (MMP-3)	5×10^3 $\text{M}^{-1}\text{s}^{-1}$	Placeholder	∘	Act. rate	None
[SLPI]	200 ng/mL	Ashcroft 2000[6]: no conc. stated	∘	HNE (n/v)	None

**Low sensitivity: even at 10× lower [A1AT] than estimated, HNE $t_{1/2} = 55$ ms; HNE non-viability conclusion unchanged.*

References

All papers in `electrochemical_device/papers/layer_00/` unless otherwise noted.

[1] Trengove NJ, Stacey MC, Macauley S et al. (1999) Analysis of the acute and chronic wound environments: the role of proteases and their inhibitors. *Wound Repair Regen* 7(6):442–52. **trengove1999.pdf** – values from text confirmed.

[2] Ladwig GP, Robson MC, Liu R et al. (2002) Ratios of activated matrix metalloproteinase-9 to tissue inhibitor of metalloproteinase-1 in wound fluids are inversely correlated with healing of pressure ulcers. *Wound Repair Regen* 10(1):26–37. **ladwig2002.pdf** – chronic concentrations from text (p.31) confirmed. Healing concentrations from Figure 8 only (approximate).

[3] Nagase H, Woessner JF Jr (1999) Matrix metalloproteinases. *J Biol Chem* 274(31):21491–4. **nagase_woessner1999.pdf** – MMP overview. Contains no kinetic constants for TIMP-1/MMP-9; cited for biological context only.

[4] Olson MW, Gervasi DC, Mobashery S, Fridman R (1997) Kinetic analysis of the binding of human matrix metalloproteinase-2 and -9 to tissue inhibitor of metalloproteinase (TIMP)-1 and TIMP-2. *J Biol Chem* 272(47):29975–83. **olson1997.pdf** – k_{on}, k_{off}, K_i for MMP-9 (82 kDa)/TIMP-1 confirmed from Table IV (enzyme inhibition assay).

[5] Beatty K, Bieth J, Travis J (1980) Kinetics of association of serine proteinases with native and oxidized α-1-proteinase inhibitor and α-1-antichymotrypsin. *J Biol Chem* 255(9):3931–34. **beatty1980.pdf** – $k_{\text{on}}(\text{HNE}/\alpha_1\text{-PI}) = 6.5 \times 10^7$ $\text{M}^{-1}\text{s}^{-1}$ confirmed from Table 1.

[6] Ashcroft GS, Lei K, Jin W et al. (2000) Secretory leukocyte protease inhibitor mediates non-redundant functions necessary for normal wound healing. *Nat Med* 6(10):1147–53. **ashcroft2000.pdf** – confirms SLPI role in wound healing (knockout mouse model). Does not provide SLPI concentration in human wound fluid;

SLPI concentration remains unconfirmed in this model.